

Institute of Actuaries of India

Subject CP1 – Actuarial Practice (Paper A)

November 2020 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

- i) Yes, as liability outgo is based on estimates –risk of model error, parameter error etc and uncertain as to timing as well
Premium inflow (renewals may be lower, new business may be lower than expected) are also based on expectations which may not materialize

Coupons/redemptions may not happen due to credit defaults/roll over of debt/options attached to debt instruments

Timing mis-match between asset and liability cash flows [3]

ii)

- Cash is most attractive asset now due to expected increase in interest rates which would make long term debt unattractive
- Equity/property/other asset choices are overpriced currently
- Great uncertainty over future economic situation

[3]

[6 Marks]**Solution 2:**i) **Short term**

- Withdrawal rates may increase as a result of bad publicity and concerns of policyholders regarding the security of their benefits [0.5]
- Agents and distributors may encourage customers to transfer their existing policies to other insurers [0.5]
- The effect of withdrawal rates is less certain if the policyholders see the prospect of any takeover by a large financially strong player life insurer [1]

Long –Term

- Some of these withdrawals may be selective, leading to worsening of mortality and sickness experience of the company. [1]
- Consumers may also be conscious about penalties over surrender and may defer/not surrender if they are nearing maturity [0.5]
- It is likely that as the service standards decline over time, withdrawals may increase. [0.5]
- It is likely that insurer could follow a prudent investment strategy in which case owners of with profit policies, investment linked products find the investment performance lagging competition and may surrender [1]

[5]ii) **Short-term**

- No new business strain will have a positive impact on the solvency position in the short term, which might suggest a less constrained investment policy. [1]

- However, the investment policy is likely still to be very constrained given the poor solvency position and possible regulatory constraints [1]

Long –Term

- In the longer term, when the liability duration becomes much shorter, the assets portfolio could be moved more towards shorter-dated, fixed –interest type investments to reduce the asset volatility. [1]
- The return volatility will increase relative to the size of fund as the size of fund reduces [0.5]
- The need for liquidity may increase as the funds runs off and if withdrawals increase. [0.5]
- The company may be forced to dispose of illiquid assets at an unfavourable time. To prevent this, it could undertake a gradual move into more liquid assets (for e.g. less direct property, more cash) [1]
- As the fund under management decrease, the dealing cost will increase relative to the size of the funds, reducing net returns [0.5]
- It will become harder to attract and retain good fund managers and investment return might be negatively affected. [0.5]
- The tax position may change as a result of contracting rather than expanding funds [0.5]

[Max 3]

[5]

[10 Marks]

Solution 3:

i) Features & Benefits

- Only cover that specific infectious disease and exclude any other treatment to manage costs
- Offer short term policies like 3 moths, covering only the monsoon period
- Reasonable Sum Insured based on observed trends of costs for managing the disease– one single SI or few levels of SI could also be offered
- Comprehensive in-patient hospitalisation cover / or
- home nursing for treatment taken at home, if hospital beds are not available
- Coverage for consultations, diagnostic tests, pharmacy and
- Coverage even for non-medical expenses incurred during hospitalisation for example medical consumables, customised diets etc.
- First 15 days waiting period from date of inception of policy to avoid anti-selection.
- No other waiting periods
- No other exclusions
- Post hospitalization cover for follow up after discharge
- Simple Options for sub limit on room rent to limit costs
- Simple extra loading stricture for customers suffering from co-morbidities

[0.5 each, **Max 5**]

ii) Reinsurance program:

Quota share

1. Since it is a small company capital usage can be optimised by Quota share agreement - share in each claim proportionally as in premium
2. Can write more business with same amount of capital [2]

XOL arrangements

3. Risk XL prevent from large one off claims but since SI could be low, such arrangement could be of limited use
4. Aggregate XoL/ CAT cover is mandatory as -Aggregation of risk: since this is a vector borne disease there is significant risk of aggregation of claims/catastrophic risk/outbreak risk [2]

Stop Loss cover

5. As an alternative to XoL, a stop loss cover could also be considered as there would be greater uncertainty about future experience
6. Here reinsurance kicks in to protect P and L in a bad flu season [1]

[5]

[10 Marks]

Solution 4: Ways to improve capital position

- 1) Increase the amount of own funds. This could be via a loan issue, share issues, group capital injection etc.

Advantages & disadvantages:

- + This will directly increase the own funds and hence the capital coverage ratio
- + This could be relatively straightforward depending on the company's capital and group structure
- This will diminish the return on capital and more profits will be required to maintain the Return on Equity hurdle
- It could be inefficient to raise more capital (Group could have other uses, loans might be available at high interest and share price might not be favourable)
- It may be difficult to raise debt/equity in this stressed economic times when there are credit defaults [3]

- 2) Increase the amount of reinsurance ceded/purchased

Advantages & disadvantages:

- +Relatively straightforward to arrange.
- +Could be done with good profit sharing and commission structures in place
- Will also cede away some of the profit
- Will need to be cede to reinsurers with good credit quality otherwise counterparty default risk will increase the Solvency Capital Ratio and the overall benefit will not be as good as anticipated
- Might not be available or only available at high price.
- This will diminish the Return on Equity achieved as reinsurers generally price in a profit margin
- Local reinsurers may also have been downgraded in stressed economic times, so may have to deal with foreign reinsurers which increases currency risk, deposits withheld etc.

[3]

- 3) Adjust the business plans and strategy to underwrite less risk in the future. I.e. negative growth

Advantages & disadvantages:

- +Will reduce the underwriting risk contribution and lead to lower capital requirement
- Will cause pressure on fixed expenses percentage contribution as there will be less premium to spread costs over
- +Could be beneficial in uncertain economic times. I.e. by sticking with stricter underwriting guidelines. If underwriting strategy is good you will remain competitive where you want to be and this could lead to a saving on claim costs.
- Might not be in line with shareholder expectations. Will want to achieve profitable year on year growth

[3]

4) Reinvest assets into less risky perceived classes

Rebalancing the asset portfolio so as to decrease the shocks applied in the standard formula in the form of:

- +Having less exposure to equities/risky debt
- +Ensuring enough diversification between counterparties

Advantages & disadvantages:

- + Will lead to lower capital requirement due to better quality portfolio
- Could lead to lower overall returns achieved, due to less perceived riskiness
- Might be difficult to sell poor quality debt but difficult if there are credit defaults-no buyer found
- lack of good debt and matching by term

[3]

[12 Marks]

Solution 5:

i)

Principles of investment for a provider of defined benefit schemes are as follows:

- To maximise the return on invested assets subject to an acceptable level of risk of failing to meet the Scheme's liabilities over the long term;
- To ensure that the Scheme meets its funding obligations ;
- To generate a return on assets that will help to stabilise and minimise the long term cost of the Scheme.

[2]

ii) In framing an Investment strategy trustees to consider:

- are invested in a manner designed to ensure the security, quality, liquidity and profitability of the portfolio as a whole so far as is appropriate having
- regard to the nature and duration of the expected liabilities of the Scheme which will generally be long term
- Investments are properly diversified in such a way as to avoid excessive reliance on any particular asset, issuer or group of undertakings and so as to avoid accumulations of risk in the portfolio as a whole;
- Ensure investments are tax efficient;
- include use of derivative instruments only in so far as they contribute to a reduction in investment risks or facilitate efficient portfolio management.
- Take into current regulatory framework,
- sponsor's risk appetite or
- any ethical constraints

[4]

iii) Cons

- 1) Stocks have fallen by 30% but can fall further more in the short term.
- 2) Do sponsor's have the risk appetite for short term volatility
- 3) There might be regulatory or self-imposed restrictions which prevents the fund from doing this and such actions can lead to regulators intervening
- 4) There will be cost involved in sale and purchase of assets
- 5) Higher costs compared to managing debt and cash
- 6) Need to build expertise or hire new managers as well
- 7) Other assets might provide tax benefits better than equities.
- 8) There might be a regulatory limit of what percent of Investments in equities will be considered for covering the fund liabilities in determining solvency of the fund

Pros

- 1) It is a DB scheme and hence liabilities would be long term and some of it could be real
- 2) Equities are best match for real liabilities in the long term
- 3) Historically equity has given higher return especially when invested in economic downturn
- 4) These are blue chip stocks as well which are currently undervalued
- 5) Equity is better asset class when inflation adjusted compared to debt and cash
- 6) Some hedging could also be considered to protect from down side risk associated with equity
- 7) Would improve diversification in the fund
- 8) Higher returns in the long term may lead to sponsor lower contribution better benefits for members

[0.5 each, Max 8]

[14 Marks]

Solution 6:

i) Data required for analysis

Analysis is required to be performed by key risk factors to understand the differences in experience across groups. [0.5]

Exposure data

- Number of electric cars sold by month/qly/annual (based on data availability) from launch of scheme to till date [1]
- Customer profile by age, location, gender, past driving experience, mileage which could be key risk factors [1]

Claims data

- Claim profile by age, gender, location, past driving experience, mileage,
- Actual maintenance cost and trends by various parameters as mentioned above
- Principal causes of claims (for eg., any specific component, mechanism)
- Average cost of claims by major causes of claim
- Break up costs by labour and material/parts and its trend across years
- Budgeted repair cost and basis for the same for comparison of actual by expected

[0.5 each, Max 4]

ii) Improving benefit design

- Reduce coverage period from 6 years to say 3 years

- Exclude repairing or maintenance cost for certain parts which would generally deteriorate due to faster wear and tear for eg., plastic parts, tyres etc
- Exclude repairs claims from accidents, negligence of owners(for eg., not charging as per mandated norms) etc.
- Add a deductible or other cost sharing options
- Introduce excess i.e repair costs upto certain percentage/amount has to be borne by customer
- Do not entertain claims from vehicle non-usage beyond say 90 days
- Do not entertain claims caused by natural disasters (flooding)

[0.5 each, Max 3]

Implications of the changes to scheme design

- Reducing the term may be a good option to reduce cost since costs escalate with duration of the plan being in force and wear and tear will not be covered
- A deductible or co-payment will immediately reduce the cost of the plan
- It may even ensure that clients look after their vehicles better and hence may reduce the cost not just from the co-payment itself but also the behavioural impact
- Excluding certain events may reduce overall cost if that is one of the reasons for escalating charges
- These options may improve the cost of the scheme without deterring clients from buying the vehicles
- This will only help for new vehicles sold. All existing clients will still claim under the scheme.

[0.5 each, Max 3]

Proposed Other Mitigation:

- **Charge for the scheme**
- **The charge could be flat rate based on expected business mix**

Implications

- Could impact sales volumes
- This will again help new sales only
- Some level of cross subsidy from risk groups to high risk groups
- Business mix could be different than expected leading to low/high costs compared to costs

[3]

• Other mitigation- Purchase insurance

Implications

- This will remove any uncertainty as the insurer will charge a premium that can be determined upfront
- It may be possible to find cover for existing clients as well
- The downside is that the insurer will add a margin for uncertainty and profit which may render the benefit rather expensive

[3]

[16 Marks]

Solution 7:

- i) Initial underwriting covers medical underwriting, life style underwriting and financial underwriting

Lives that present themselves for insurance are from a heterogeneous group and the purpose of underwriting is to classify individuals into relatively homogenous groups and charge appropriate risk premium.

This is done through

Medical underwriting- assess the health of an individual ;

Life style underwriting:- assess impact of an individual's life style (for eg., occupation, leisure activities on risk)

This enables a company to identify lives which could be offered "standard terms" and those to whom special terms have to be offered which could include-

- Charging an extra premium
- Declining the application
- Postponing the application
- Introducing specific exclusions etc.

Financial underwriting: to avoid over insurance and moral hazard

[5]

ii) Advantages:

- reducing underwriting can reduce expenses e.g. through a lower number of underwriters or fewer medical examinations.
- Reducing underwriting can also result in increased volumes of business sold.
- This is because distributors (and customers) may favour a more relaxed underwriting stance with less "hassle" factor (or intrusion) and reduced processing time.
- Lower costs could lead to lower and therefore more attractive premiums (all else being equal) which may outweigh any increase in morbidity cost.
- Greater sales volumes could increase total profits to the company and reduce per policy expenses, but may lead to admin/capital strain.
- There will also be the implementation costs for any new processes introduced, including any systems changes, amendments to the wording of application forms and proposals and retraining /redundancy costs.
- The company also needs to consider higher potential claims costs, the risk of increased anti selection and lapse and re-entry.

[1 each, Max 4]

Disadvantages

- If the company adopts "accept/decline" strategy given the reduced underwriting information, then more customers may be declined than previously.
- This could adversely impact the company's reputation, which could impact sales volumes.
- Reducing initial underwriting is likely to lead to increased claims underwriting and hence more claims being denied which would harm reputation of company.
- Reduced financial underwriting may lead to moral hazard.
- If reducing the level of underwriting would put the insurer out of line with its competitors then the insurer is likely to attract a disproportionate share of the adverse risks (i.e. anti-selection), which would increase the average morbidity experience of the portfolio.
- The insurer will then want to charge for this additional morbidity risk and so is likely to have to increase premium rates.
- If premium rates are increased then this is likely to be unpopular with the insurer's sales channels, which could offset the sales advantage from the reduction in the level of underwriting.
- But this can further exacerbate the anti-selection effect, as the healthier lives are more likely to be take advantage of lower basic premium rates offered elsewhere where there is stricter underwriting.
- Further, if reinsurance is used then it is likely that the reinsurer will either increase its rates significantly if the level of underwriting is significantly reduced or in the extreme may not wish to

continue reinsuring this product and this might also be reflected in higher premiums (or lower profit margin).

- The company may also need to hold higher reserves due to increased uncertainty about future morbidity experience. Increased reserves may affect the solvency of the company.

[1 each, Max 7]

[16 Marks]

Solution 8:

i) Checks to be applied on Claims data

- Claims ID should not be missing
- Mapping of Policy Number and claims -else error record
- Dates missing, Missing data fields, Duplicate entries
- Date of accident is within policy period and after policy commencement date
- Date of intimation is before valuation date and after accident date/ policy commencement date
- Claims status appropriate– settled , outstanding, refuted, reopened and not blank
- Amounts settled/outstanding not blank or 0
- Claim type field not blank- injury, death, property damage
- Direct claims related expense not 0
- If quota share agreements in place-reinsurance recoveries not blank
- Movement of claims data from last period –changes are appropriate and ideally last triangulations results should be able to be recalculated from current data
- Reconciliation with financial statements of outstanding claims, paid claims, claims expenses

[5]

ii) In favour

Large claims should ideally be analysed separately as they normally tend to have a different frequency/severity as compared to small/normal claims and [1]

They tend to have a different development pattern than normal claims. [1]

If we continue triangulation without any adjustment, large claims would dominate the experience of the group [1]

not excluding them can mean that an “all claims” analysis shows unstable experience due to distortion from those few large claims [1]

Against

- Motor liability portfolio may result in some very large claims. However, it would not be expected that a single large claim or a few large claims would influence the reserving/triangulation results so much if the claims per se are generally large in number. [1]
- However, having a limited number of large claims might mean there is little data on which to base a separate allowance for large claims but [1]

Hence finally.

- If the large claims are expected to be completely one-off (for eg., large judicial settlements), and not likely to be repeated in the future, exclude them altogether from the data and account for them elsewhere- for eg., separate large claims reserves. [1]
 - If the claims are likely to occur occasionally, include only part of the cost of such claims, corresponding to the probability of their recurrence and the length of the periods being analysed and projected. (i.e. return period) [1]
- [8]**

iii)

- Consider the definition of a large claim
For example:
 - Single claim amount greater than certain value
 - Single claim amount greater than certain percentage of total capital of the company
 - Single claim amount greater than certain percentage of Gross premium of that product/Line of Business/Total GWP of the company [1.5]
 - A large claims threshold may be held constant or indexed over time [0.5]
 - Where no indexation adjustment is applied, one needs to be aware that in real terms the pool of large claims should be expected to change over time. [0.5]
 - the large threshold could be so chosen to mirror the reinsurance retention [0.5]
- [3]**

[16 Marks]
